

System Requirements Specification

Document 1 — Clean Format (Simple)

This document contains 100 system requirements organised across six levels of hierarchical section numbering. Format: plain text, uniform font, no headers or footers. Intended for pdfplumber parsing workflow testing.

1 The system shall provide a web-based user interface accessible via standard browsers including Chrome, Firefox, Edge, and Safari.

1.1 The system shall support user authentication using username and password credentials stored in the central identity provider.

1.2 The system shall enforce a session timeout of 30 minutes of inactivity, after which the user shall be required to re-authenticate.

1.3 The system shall display the currently authenticated username and role in the application header at all times.

1.1.1 The system shall validate that the password contains a minimum of 12 characters including at least one uppercase letter, one digit, and one special character.

1.1.2 The system shall lock the user account after five consecutive failed authentication attempts within a 10-minute window.

1.1.3 The system shall send an email notification to the registered address when an account lockout event occurs.

1.2.1 The system shall display a countdown warning 5 minutes before the session timeout threshold is reached.

1.2.2 The system shall provide a 'Stay Logged In' button that resets the session timer when clicked.

1.3.1 The system shall display the user's assigned role using the role name as defined in the identity provider group membership.

1.1.1.1 The system shall reject passwords that match any of the user's previous 12 passwords.

1.1.1.2 The system shall reject passwords that contain the user's first name, last name, or username as a substring.

1.1.1.3 The system shall enforce password expiry after 90 days and prompt the user to set a new password upon next login.

1.1.2.1 The system shall record each failed authentication attempt in the audit log with timestamp, source IP address, and username.

1.1.2.2 The system shall require administrator action or a verified password reset email to unlock a locked account.

1.1.3.1 The email notification shall include the timestamp of the lockout event and the source IP address of the last failed attempt.

1.2.1.1 The countdown warning shall appear as a modal dialog that does not obstruct the user's current work.

1.2.2.1 The 'Stay Logged In' action shall be logged in the session audit trail with the timestamp of the extension.

1.1.1.1.1 The password history check shall be performed server-side before the new password hash is stored.

1.1.1.1.2 The system shall store only the hash of previous passwords using bcrypt with a minimum cost factor of 12.

1.1.1.2.1 The substring check shall be case-insensitive and shall apply to both the full name and each component of a hyphenated name.

1.1.1.3.1 The password expiry notification shall be sent by email 14 days, 7 days, and 1 day before the expiry date.

1.1.2.1.1 The audit log entry for failed authentication shall be written before the response is returned to the client.

1.1.2.2.1 The password reset email shall contain a single-use token that expires after 60 minutes.

1.1.1.1.1.1 The bcrypt cost factor shall be configurable by the system administrator via the security configuration interface without requiring a system restart.

1.1.1.1.1.2 The system shall verify that the bcrypt cost factor results in a hash computation time of at least 250ms on the target server hardware before accepting a configuration change.

1.1.1.2.1.1 The substring matching algorithm shall normalise Unicode characters to NFC form before performing the comparison.

1.1.1.3.1.1 Password expiry notification emails shall be sent by the background notification service within a 1-hour window of the scheduled notification time.

2 The system shall provide a requirements management module that supports the creation, editing, and deletion of requirements artefacts.

2.1 The system shall allow authorised users to create new requirements using a structured form with mandatory and optional fields.

2.2 The system shall support versioning of requirements such that each modification creates a new version and the full change history is retained.

2.3 The system shall provide a search interface that allows users to find requirements by keyword, attribute value, or section number.

2.1.1 The requirements creation form shall include the fields: Title, Description, Priority, Verification Method, and Status.

2.1.2 The system shall generate a unique identifier for each new requirement in the format REQ-NNNN where NNNN is a zero-padded integer.

2.1.3 The system shall record the creating user and creation timestamp as immutable attributes of the requirement.

2.2.1 Each requirement version shall store the complete attribute set, the modifying user, and the modification timestamp.

2.2.2 The system shall provide a side-by-side diff view comparing any two versions of a requirement.

2.2.3 The system shall allow users to add a change justification comment when saving a modified requirement.

2.3.1 The keyword search shall search within the Title and Description fields and shall support partial word matching.

2.3.2 The attribute search shall support filtering by Priority, Status, Verification Method, and Assigned Owner.

2.1.1.1 The Priority field shall be an enumeration with values: Critical, High, Medium, Low.

2.1.1.2 The Verification Method field shall be an enumeration with values: Test, Analysis, Inspection, Demonstration.

2.1.1.3 The Status field shall be an enumeration with values: Draft, Under Review, Approved, Deprecated.

2.1.2.1 The unique identifier sequence shall be maintained per project and shall not be reused even if a requirement is deleted.

2.2.1.1 The version history shall be accessible from the requirement detail view via a 'Version History' tab.

2.2.2.1 The diff view shall highlight added text in green and removed text in red with strikethrough formatting.

2.1.1.1.1 The system shall prevent the user from saving a requirement with a Verification Method of 'Test' if Status is 'Approved' and no test case is linked.

2.1.1.2.1 Changes to the Verification Method after a requirement is in 'Approved' status shall require a second user to confirm the change.

2.1.1.1.1.1 The blocking rule described in 2.1.1.1.1 shall be configurable per project by the project administrator.

2.1.1.1.1.2 When the blocking rule prevents saving, the system shall display a specific error message identifying the missing test case link.

3 The system shall provide a traceability management module enabling bidirectional links between requirements and other engineering artefacts.

3.1 The system shall support the creation of typed traceability links between requirements artefacts and test cases.

3.2 The system shall support the creation of typed traceability links between requirements artefacts and design artefacts.

3.3 The system shall generate a Requirements Traceability Matrix (RTM) report covering all requirements in a selected module.

3.1.1 The supported link types from requirement to test case shall include: Validates, Partially Validates, and Excludes.

3.1.2 The system shall display the verification status of all linked test cases in the requirement detail view.

3.2.1 The supported link types from requirement to design artefact shall include: Satisfies, Derived From, and Conflicts With.

3.3.1 The RTM report shall include columns for: Requirement ID, Title, Priority, Verification Method, Linked Test Cases, and Test Status.

3.3.2 The RTM report shall be exportable in CSV and PDF formats.

3.1.1.1 The system shall mark a link as Suspect when the source requirement is modified after the link is created.

3.1.1.2 The system shall notify the link owner by email when a Suspect flag is set on a link they own.

3.1.2.1 The verification status summary shall show counts of Passed, Failed, Not Run, and Blocked test cases linked to the requirement.

3.3.1.1 The RTM shall include a summary row at the end showing total requirement count and percentage with full test coverage.

3.1.1.1.1 The Suspect flag shall persist until a user explicitly clears it after reviewing the link validity.

3.1.1.1.2 The system shall record the user and timestamp when a Suspect flag is cleared in the link audit trail.

3.1.1.2.1 The Suspect notification email shall include a direct link to the affected traceability link in the system.

3.1.1.1.1.1 Clearing a Suspect flag shall require the user to enter a justification comment of at least 20 characters.

3.1.1.1.1.2 The justification comment for Suspect flag clearance shall be visible in the link audit trail without requiring special permissions.

4 The system shall provide a baseline management module supporting the creation and retrieval of immutable requirement snapshots.

4.1 The system shall allow project administrators to create a named baseline of all requirements in a module at any point in time.

4.2 The system shall prevent modification of any requirement artefact that is part of a locked baseline.

4.3 The system shall provide a baseline comparison view showing differences between two selected baselines of the same module.

4.1.1 The baseline creation process shall capture the complete attribute set of every requirement in the module at the moment of creation.

4.1.2 Each baseline shall be assigned a unique name, a version number, and a creation timestamp.

4.2.1 An attempt to modify a requirement in a locked baseline shall return an error message identifying the baseline name and the locked requirement ID.

4.3.1 The baseline comparison shall highlight requirements that were added, removed, or modified between the two selected baselines.

4.3.2 The comparison view shall allow the user to filter to show only added, only removed, or only modified requirements.

4.1.1.1 The baseline creation shall be atomic — either all requirements are captured or none are, with no partial baseline state.

4.1.2.1 The version number shall follow semantic versioning format MAJOR.MINOR where MAJOR increments on release baselines and MINOR on interim baselines.

4.3.1.1 For modified requirements, the comparison shall show the changed attribute values with old and new values presented side-by-side.

4.1.1.1.1 If baseline creation fails due to a system error, the system shall log the failure, notify the administrator, and ensure no partial baseline record is stored.

4.1.2.1.1 The system shall prevent the creation of a new baseline with the same name as an existing baseline in the same project.

4.1.1.1.1.1 The atomicity guarantee shall be implemented using a database transaction that is rolled back in its entirety on any failure.

5 The system shall provide an import module supporting the ingestion of requirements from external document formats.

5.1 The system shall support the import of requirements from Microsoft Word documents in .docx format.

5.2 The system shall support the import of requirements from CSV files conforming to the project's defined import template.

5.3 The system shall support the import of requirements from ReqIF packages in .reqif and .reqifz formats.

5.1.1 The Word import shall recognise Heading 1 through Heading 6 styles as hierarchy level indicators.

5.1.2 The Word import shall create one requirement artefact per paragraph that is not formatted as a Heading style.

5.2.1 The CSV import template shall require the columns: Identifier, Artifact Type, Name, Primary Text, and parentBinding.

5.2.2 The system shall validate the CSV file against the template before commencing import and shall report all validation errors before processing any row.

5.3.1 The ReqIF import shall preserve all custom attributes defined in the ReqIF specification file.

5.1.1.1 A Heading 1 paragraph shall create a Level 1 requirement artefact; a Heading 2 shall create a Level 2 child of the nearest preceding Level 1, and so on.

5.2.1.1 The parentBinding column shall reference the Identifier value of the parent requirement within the same import file.

5.2.2.1 Validation errors shall be reported in a downloadable error report CSV with columns: Row Number, Column Name, Error Description.

5.1.1.1.1 If a Heading level is skipped (e.g., Heading 1 followed directly by Heading 3), the system shall insert a placeholder Level 2 artefact and flag it for review.

5.2.1.1.1 A parentBinding value that references an Identifier not present in the same import file shall be treated as a validation error.

5.1.1.1.1.1 Placeholder artefacts created for skipped heading levels shall be assigned the name '[AUTO-PLACEHOLDER]' and the status 'Requires Review'.

5.1.1.1.1.2 The system shall generate a post-import report listing all placeholder artefacts created during the import process with their location in the module hierarchy.